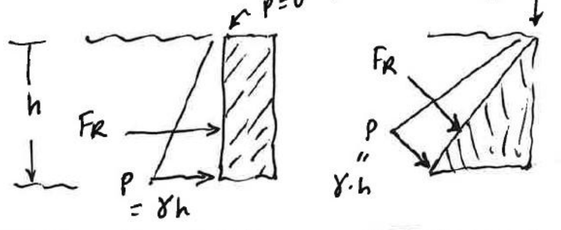
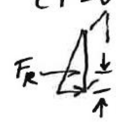
Fluid Pressure
Useful Area and Centroid Formulas

1. Flat Plate, Constant Width, extends to (free) surface: Triangular Distribution

2. Flat Plates



(p=0 at surface)

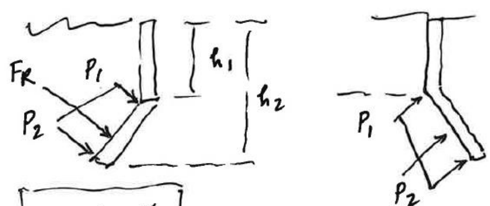


Triangular Pressure Distrib ⊥ to plate.

$$\boxed{F_R = \tfrac{1}{2} b h \cdot (\text{width of plate})}$$

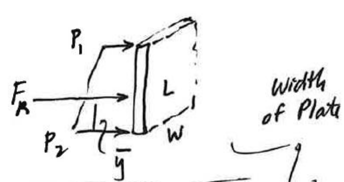
2. Flat Plate, Constant Width, does NOT extend to (free) surface: Trapezoidal Distribution

3. Flat Plates (at depth)



$$\boxed{\begin{aligned} P_1 &= h_1 \cdot \gamma \\ P_2 &= h_2 \cdot \gamma \end{aligned}}$$

Trapezoid



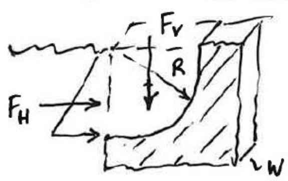
$$\boxed{F_R = \left(\frac{P_1 + P_2}{2}\right)(L)(w)}$$

$$\boxed{\bar{y} = \frac{1}{3}\left[\frac{2P_1 + P_2}{P_1 + P_2}\right] L}$$

3. General Curved Surface, Constant Width: Must use $F_R = F_H + F_V$

(a) Quarter Circle

Examples: 1/4 circle



Δ for F_H

wt of fluid for F_V

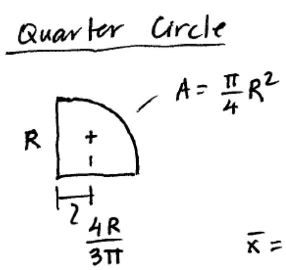
$wt = VOL \cdot \gamma = F_V$

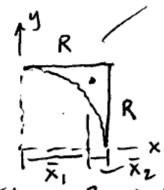
$$F_V = \underbrace{\left(\frac{\pi}{4} R^2\right)}_{\text{Area}} (w) \cdot \gamma$$

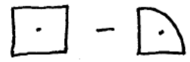
Acts at centroid:

$$\bar{x} = \frac{4R}{3\pi}$$

(b) Quarter Circular Spandrel (the roughly triangular area between a curve and a vertical line)







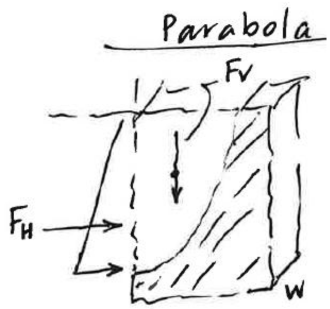
$$A = R^2 - \frac{\pi}{4}R^2 = (1 - \pi/4)R^2$$

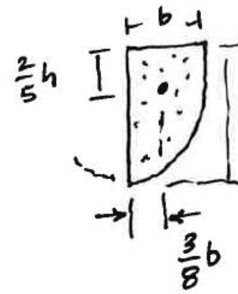
$$\boxed{A = 0.2146\, R^2}$$

$$\bar{x}_2 = (1 - .7766)R = .2234\,R$$

$$\bar{x} = \frac{\Sigma \tilde{x} A}{A_T} = \frac{R/2\,(R^2) - \frac{4R}{3\pi}\left(\frac{\pi}{4}R^2\right)}{A_T} = \frac{1/6\, R^3}{.2146\, R^2} = \boxed{.7766\,R = \bar{x}}$$

(c) Parabola (Area = 2/3 b*h; Volume = 2/3 b*h*width)





(d) Parabolic Spandrel (the roughly triangular area between a curve and a vertical line)
Area = 1/3 b*h; Volume = 1/3 b*h*width